Visualizing BAM files.

Always visualize your data

Biological data is very rich in information.

No tool can describe and capture this variety.

Algorithms are good at churning through and finding well defined/expected features.

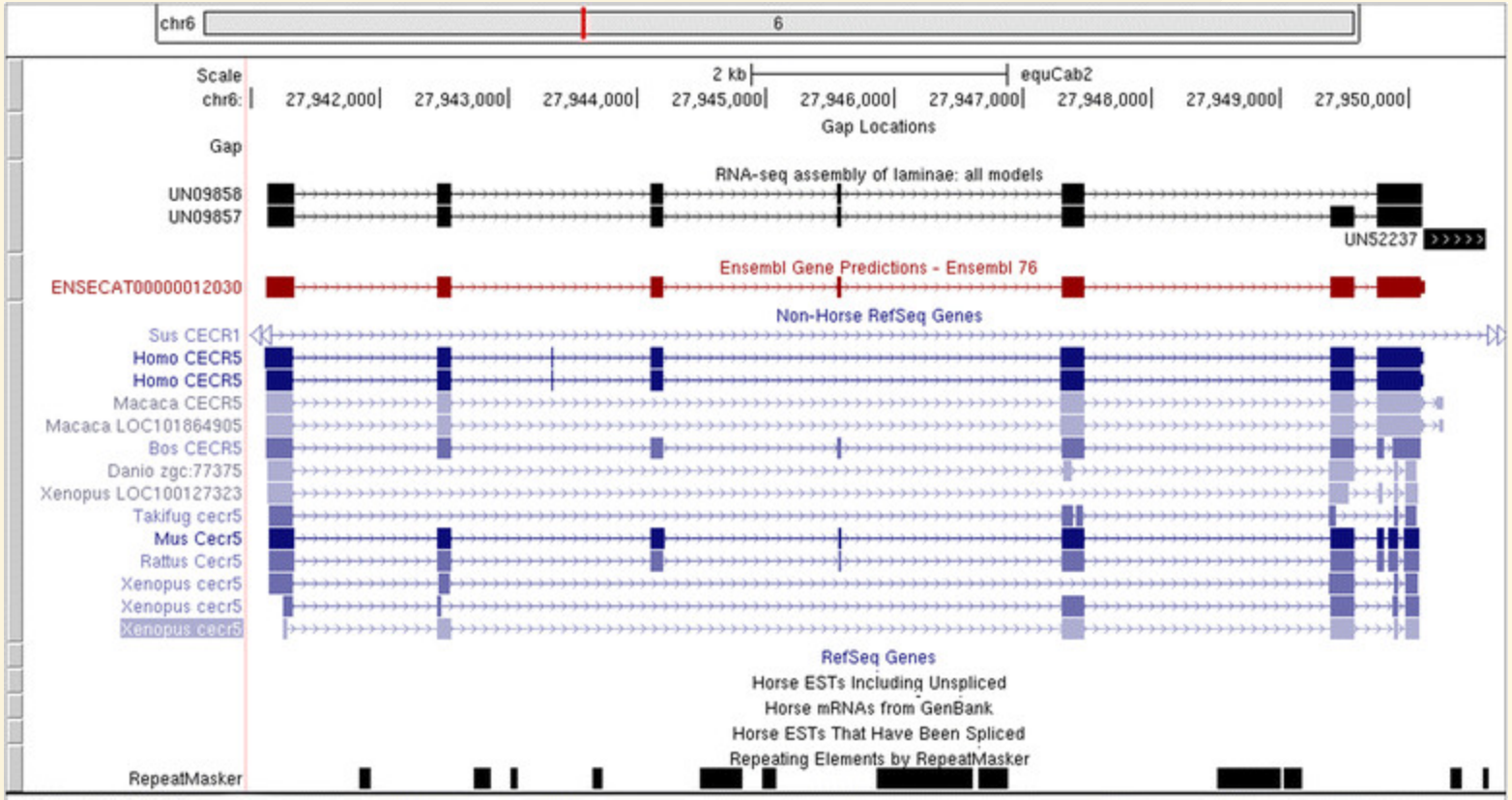
The human eye is the best instrument to see unexpected properties.

Advances in genome visualization

In year 2000 computer games looked like this



In year 2000 genome visualization looked like this



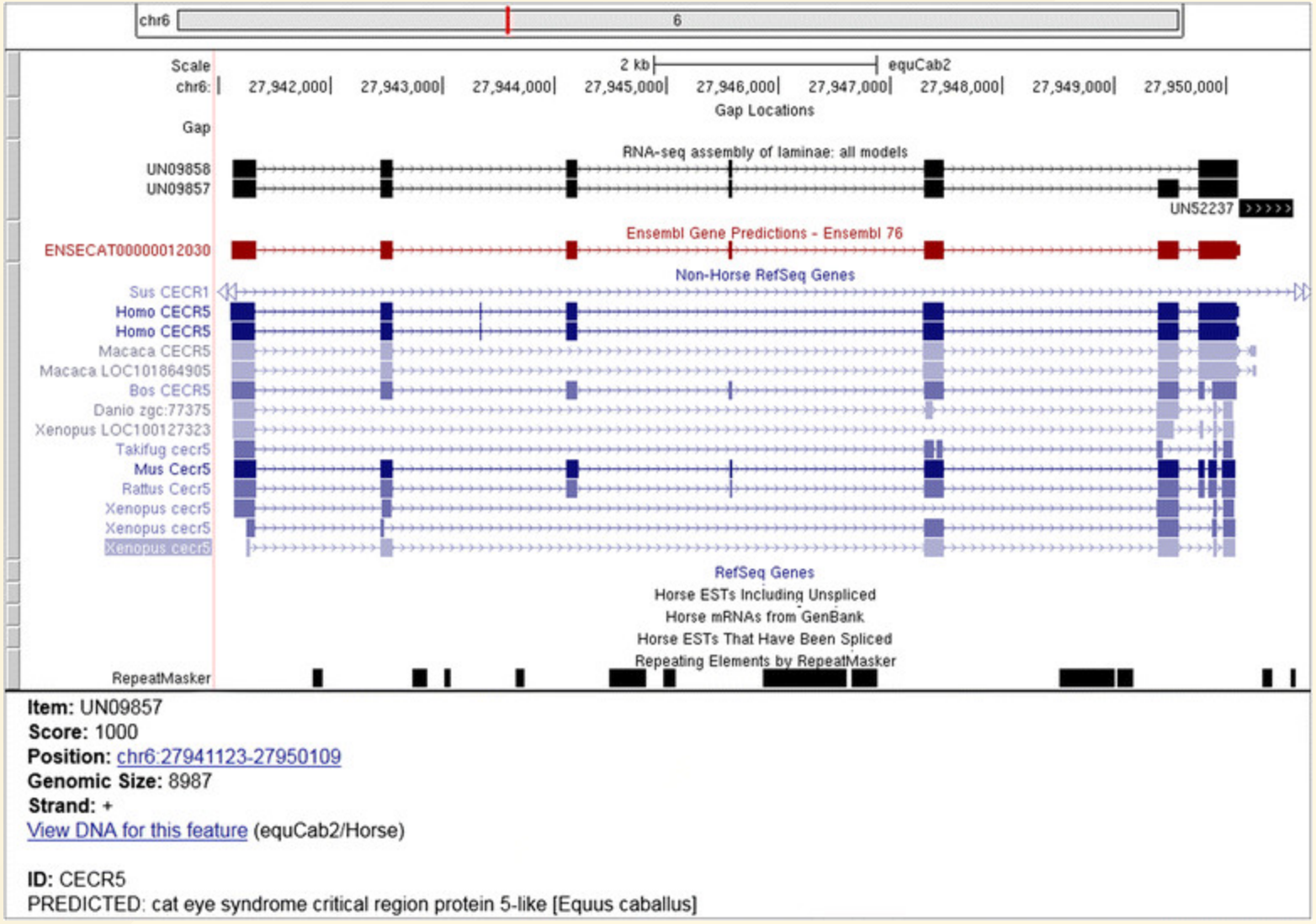
Item: UN09857
Score: 1000
Position: chr6:27941123-27950109
Genomic Size: 8987
Strand: +
View DNA for this feature (equCab2/Horse)

ID: CECR5
PREDICTED: cat eye syndrome critical region protein 5-like [Equus caballus]

By 2020 computer games look like this



By 2020 genome browsers looke like this



It has been painful to watch the lack of innovation when it comes to genome visualization.

Why do you think game graphics advances so much all the while scientific visualization stagnates?

Genome Browsers

- Can be web-based data repositories **UCSC Genome Browser**, **Ensembl Genome Browser**, **NCBI Genome Browser**
- Downloadable applications with graphical user interface with data sources: **IGV**, **IGB**
- BAM file viewers: **BamView**, **Savant**, **Tablet**, **GenoViewer**, **MochiView**, **SeqMonk**, **inGAP** ...
- Installable web applications: **Anno-J**, **JBrowse**

A surprising number of genome browsers don't work properly - full of quirks and oddities.

Scientists routinely underestimate how difficult it is to build a useful visualizer.

Which one is the best browser?

All of them a bad. Some are even worse.

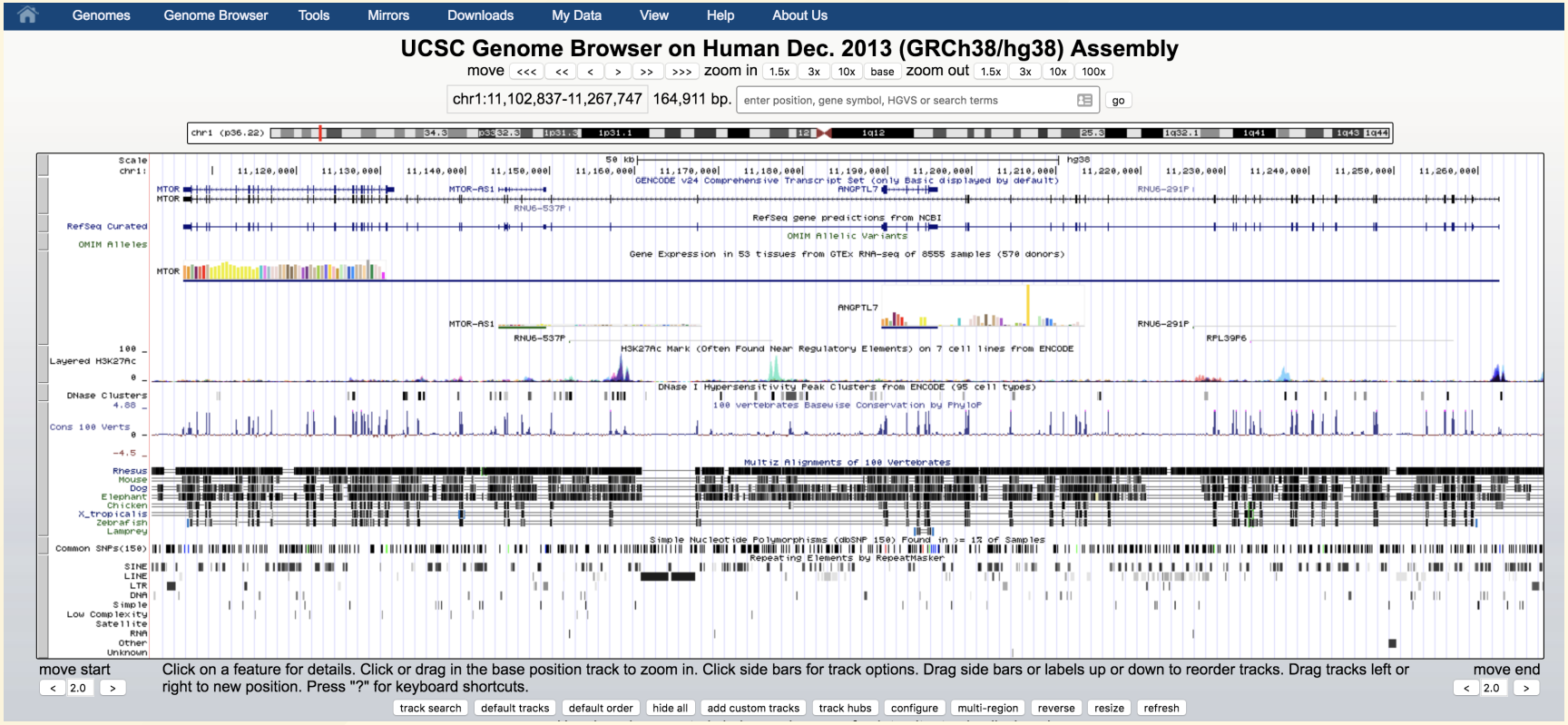
There are probably hundreds of very similar applications with various features/applications – each one is *the better* as long as you define *better* in a specific way

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Genomic data visualization is a surprisingly complex matter – users' needs diverge dramatically and can be mutually exclusive

Default view of the UCSC genome browser

Humane genome view of the UCSC genome browser.



Oh look: the elephant genome is displayed by default. Gee thanks 😬

Targeted use cases

Tools developed in a lab tend to suit the tasks performed in that environment:

- Variation for high throughput data: **IGV**, **IGB**
- Generic visualizer for genome assembly: **Artemis**

Targeted use cases:

- ChipSEQ -> **MochiView**
- DNA Methylation -> **ChipMonk** and **SeqMonk**

What gets visualized?

1. **Horizontal spans** (shown as intervals): gene locations, alignments, etc.

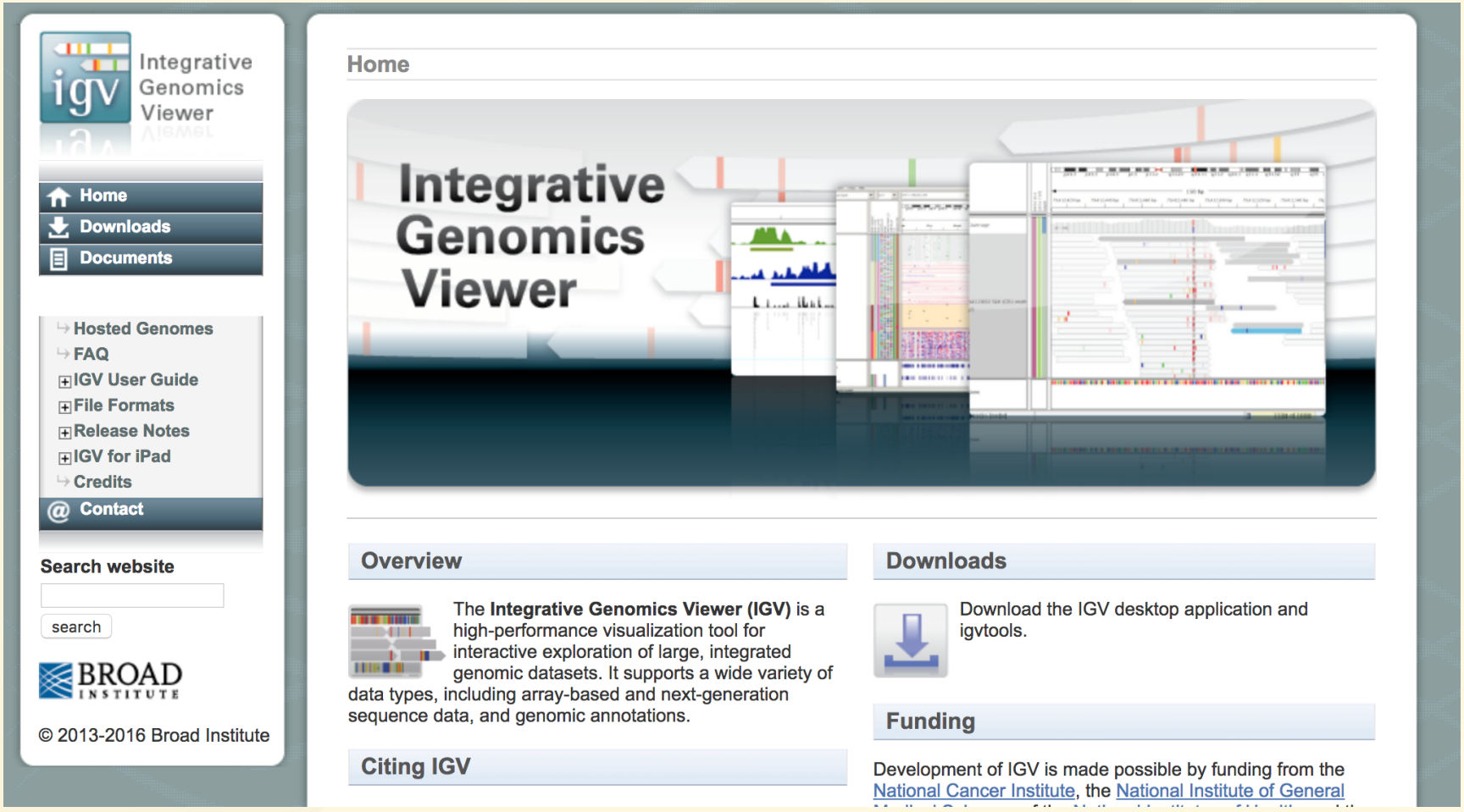
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2. **Values over intervals** (shown as vertical bars): coverages, probabilities, abundances, etc.

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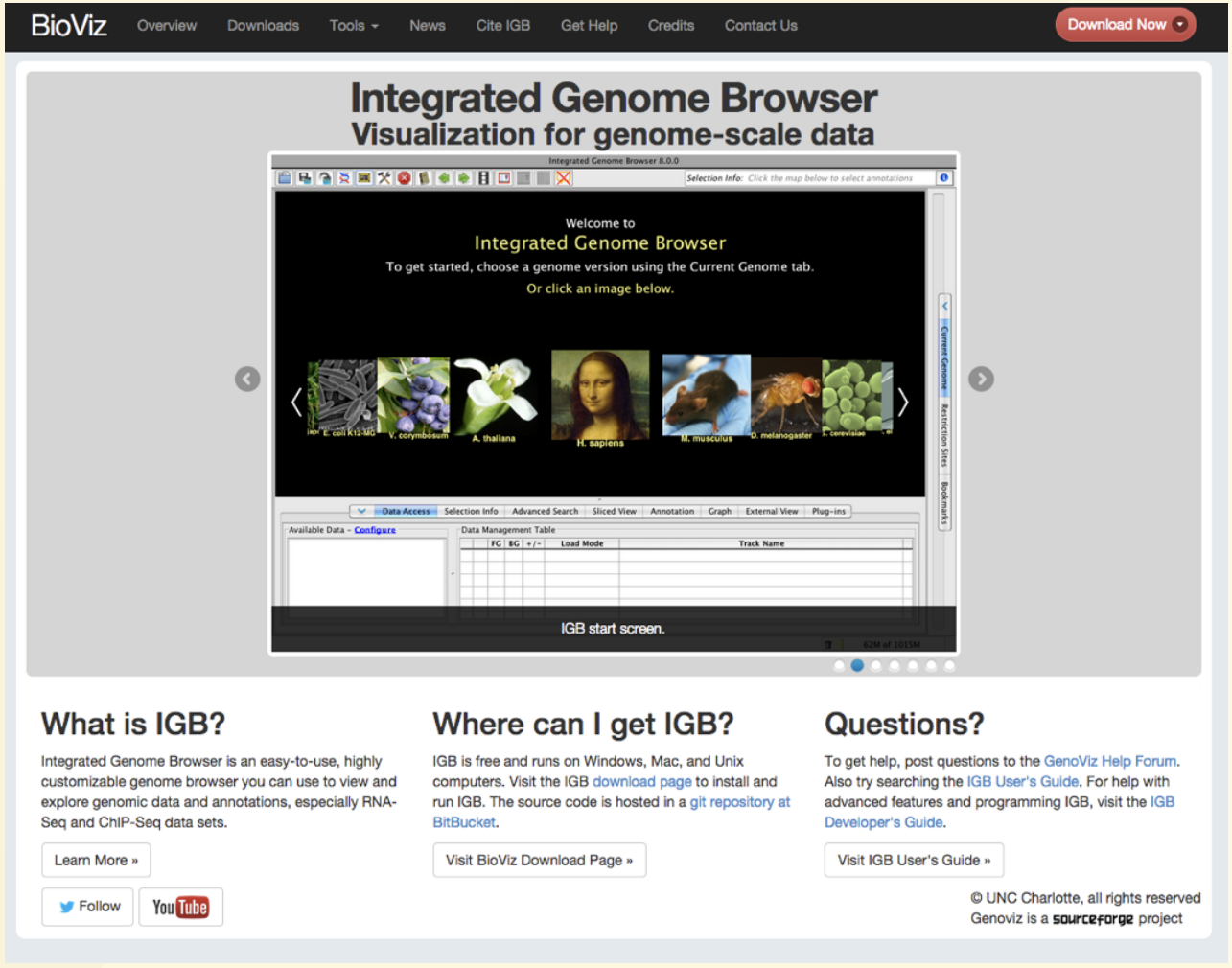
3. **Attributes** (shown as colors or "glyphs"): mutations, junctions, fusions etc.

Integrative Genomics Viewer



Developed by the Broad Institute – focus on genetic variation studies

IGB (Ig-Bee) Integrated Genome Browser



More analytics features compared to IGV.

Windows Bash Users

Install the **Windows Version** of IGV

Here is when running the files so that they are visible from Windows is important.

You need to be able to access the BAM files from Windows.

See the Windows Setup if you missed this so far

Lets make a BAM file: Get the data.

We are repeating prior steps. Keep these in scripts. Prepare the data first.

```
# The name of our reference
REF=db/ebola.fa

# Create a directory for the indices.
mkdir -p db

# Get the ebola genome
efetch -db nuccore -format fasta -id AF086833 > $REF

# Index the ebola genome
bwa index $REF

# Get the data
fastq-dump -X 10000 --split-files SRR1972739
```

Lets make a BAM file: Produce the alignments.

```
# This makes the command line more generic.
R1=SRR1972739_1.fastq
R2=SRR1972739_2.fastq


# Perform the alignment.
bwa mem $REF $R1 $R2 | samtools sort > bwa.bam

# Index the BAM file.
samtools index bwa.bam
```

Running IGV

I like to run it from the command line, but other graphical installers are also available.

I unzip the "Binary Distribution archive," move the resulting folder to `~src` then run it with:

```
bash ~/src/IGV_2.4.3/igv.sh
```

I found that the other versions are less robust when errors occur.

It is also easier to stop running if it hangs with `CTRL+C`.

How to visualize a custom genome?

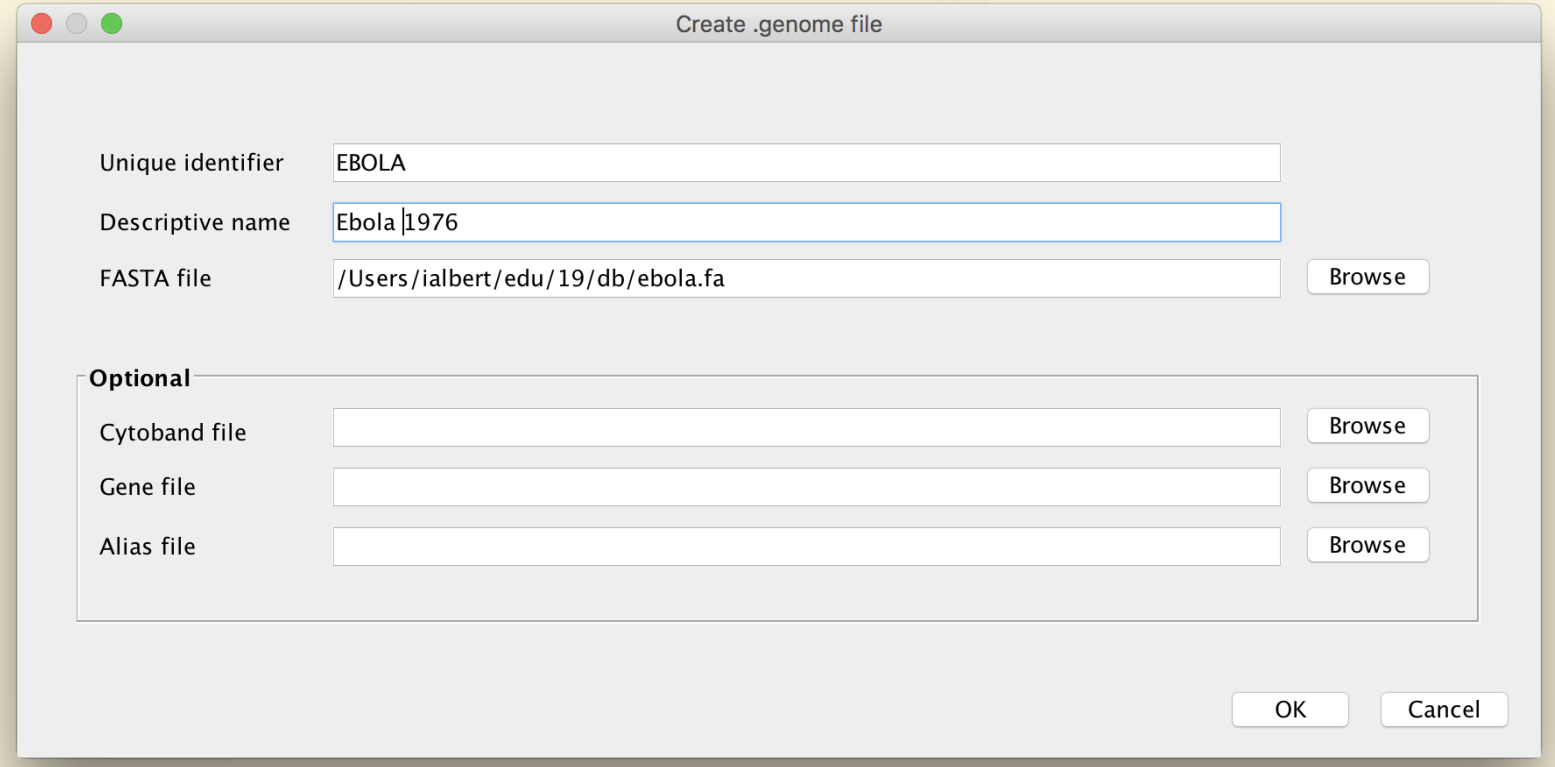
Sequence and other information for model organisms may be “pre-filled.”

Custom or less common type of data will need to be loaded manually (we will do this)

Import your genome if you are not using a standardized genome build

Import your genome

```
Menu --> Genomes --> Create .genome file
```

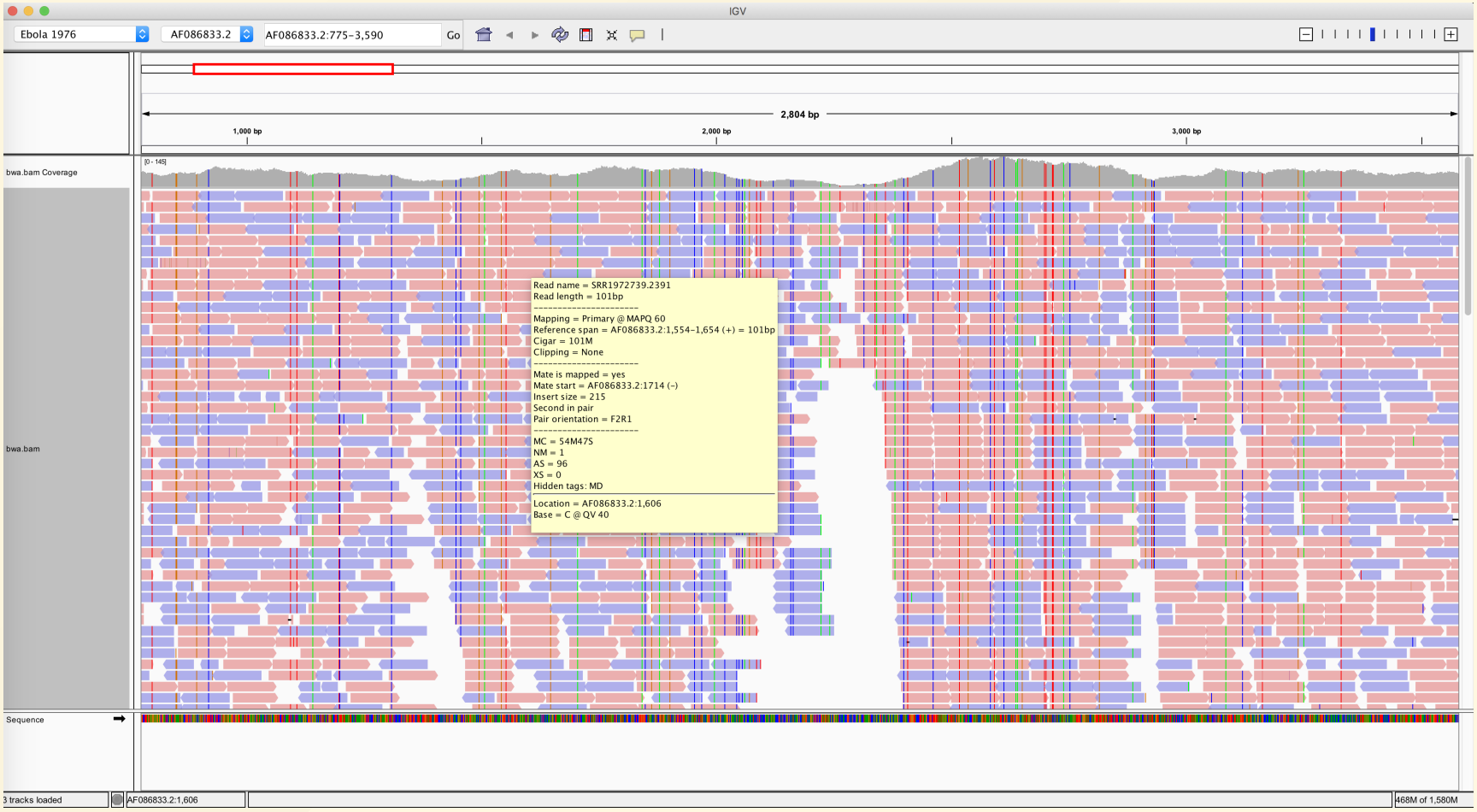


Visualize your BAM file



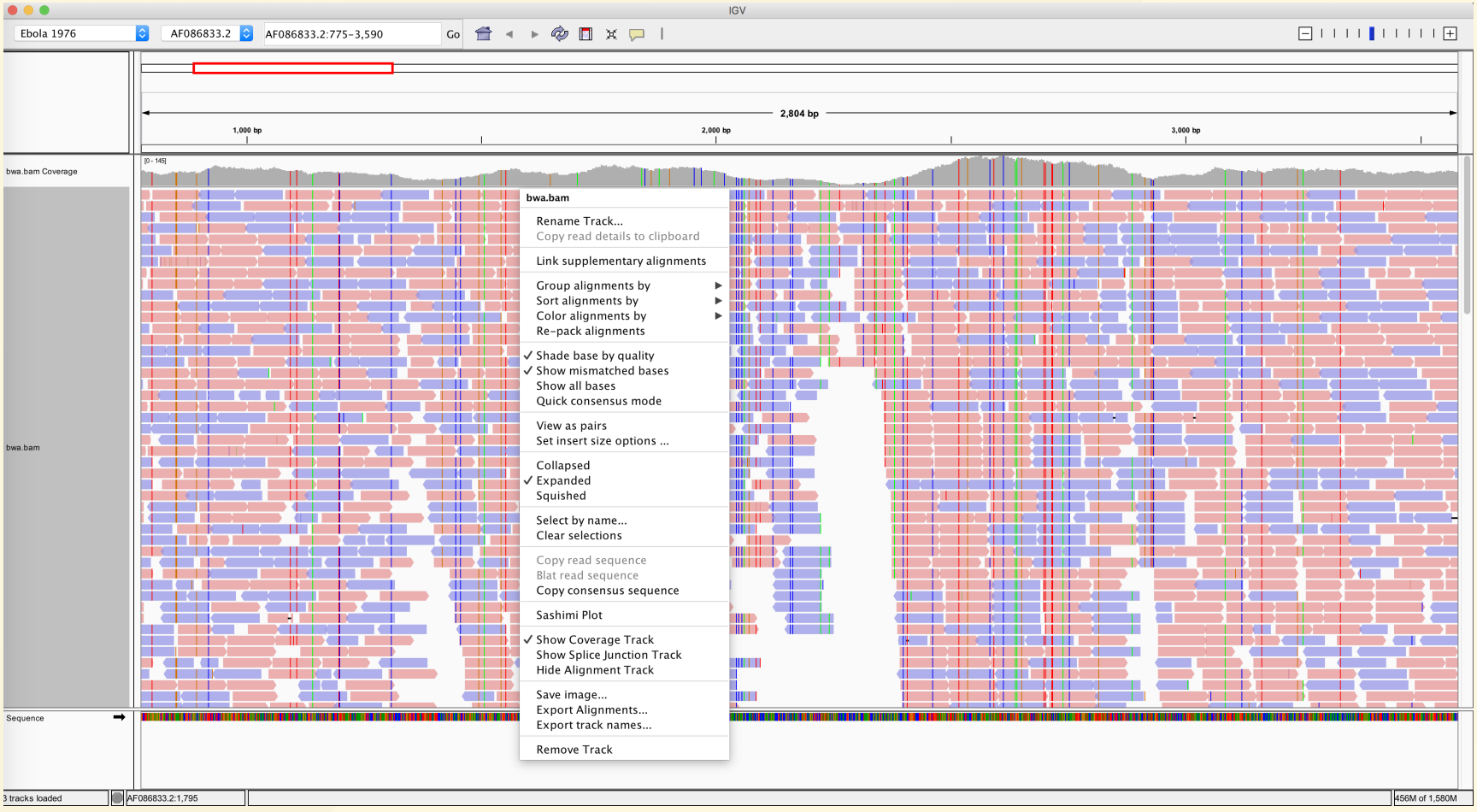
Load up your file, navigate and explore the content.

Hover to see the BAM data



Hover shows you the content of the BAM alignment.

Right click to change the visualization options



There are many choices. Explore them.